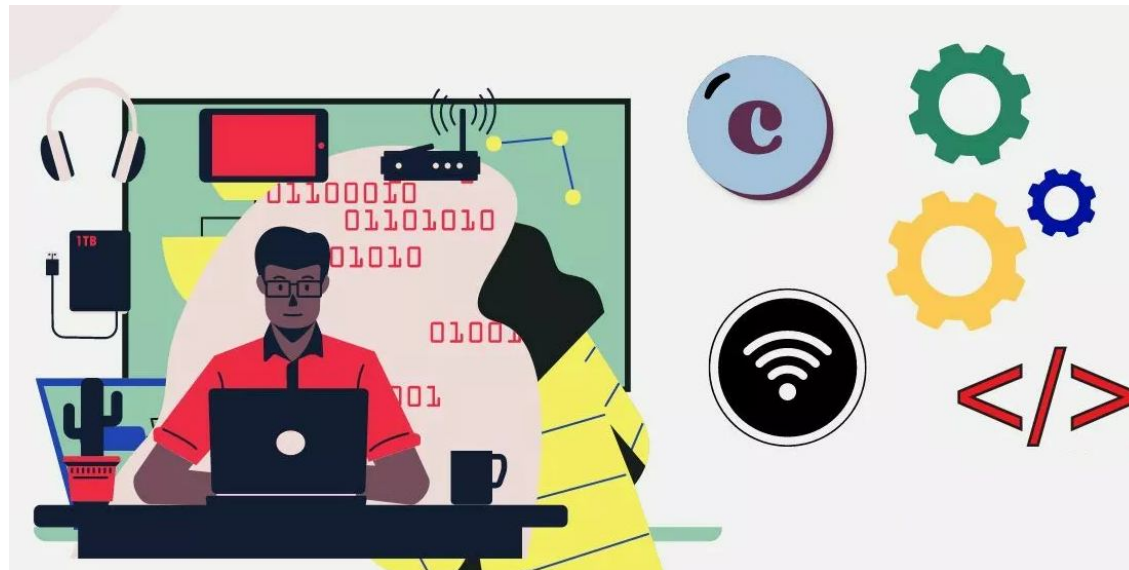




Structured Programming Concepts

(26CSE0105)



Data Types in C

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Topics to be covered

- ***Data Types in C***
- ***Basic Types:***
 - ***Integer***
 - ***Character***
 - ***Float, Double***
- ***Modifiers:***
 - ***Signed***
 - ***Unsigned***
 - ***Short, Long***

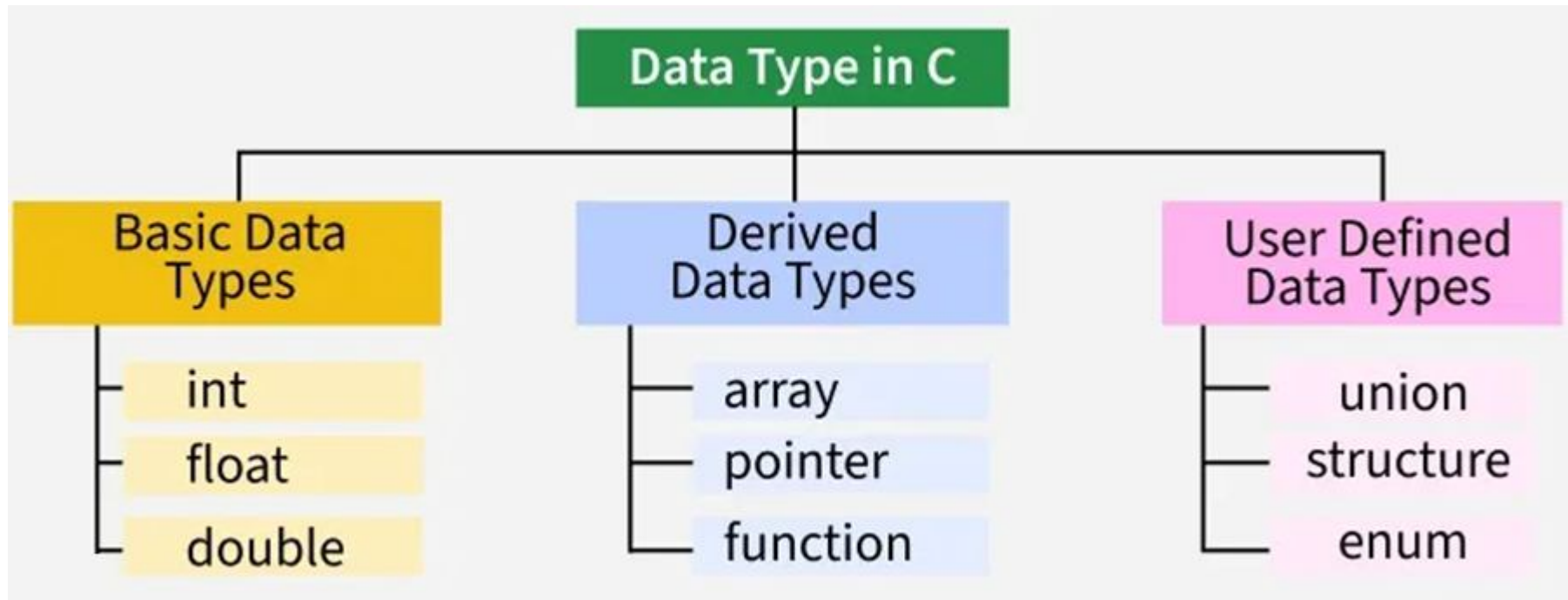
Data Types in C



- Each **variable in C** has an **associated data type**.
- It specifies the **type of data** that the variable can store like **integer, character, floating, double**, etc.
- C is a **statically type language** where each variable's type **must be specified** at the declaration and **once specified**, it **cannot be changed**.



Data Types in C



Data Types in C

Sr. No.	Format Specifier	Type	Description
1	%d	int/signed int	used for I/O signed integer value
2	%c	char	Used for I/O character value
3	%f	float	Used for I/O decimal value
4	%s	string	Used for I/O string/group of characters
5	%ld	long int	Used for I/O long signed integer value
6	%u	unsigned int	Used for I/O unsigned integer value
7	%i	unsigned int	used for the I/O integer value
8	%lf	double	Used for I/O fractional or floating data

Data Types in C

Integer Data Type:

- ❖ Stores **whole numbers** (positive, negative, or zero).
- ❖ Use **int** to declare the **integer variable**.
- ❖ **Size: 4 bytes**
- ❖ **Range: -2,147,483,648 to 2,147,483,647.**
- ❖ **Format specifier: %d**

Character Data Type:

- ❖ Stores a **single character** (like 'A', 'b', '5', or ASCII values).
- ❖ **Size: 1 byte**
- ❖ **Range: -128 to 127** (signed by default).
- ❖ **Format specifier: %c**

Data Types in C

Float Data Type

- ❖ Stores **decimal numbers** (*numbers with **fractional part***).
- ❖ **Size: 4 bytes**
- ❖ **Approximate range: $3.4e-38$ to $3.4e+38$.**
- ❖ **Format specifier: %f.**
- ❖ Sufficient for storing **6-7 decimal digits**.

Double Data Type

- ❖ Stores **decimal numbers** with **more precision** than float.
- ❖ **Size: 8 bytes**
- ❖ **Approximate range: $1.7e-308$ to $1.7e+308$.**
- ❖ **Format specifier: %lf.**
- ❖ Sufficient for storing **15 decimal digits**.

Size of Data Types in C

- The **size of data types** in **C** is dependent on the **size of architecture**.
- So we **cannot define** the **universal size** of the data types.
- the **sizeof()** operator is **used to check** the size of the data types.

```
int main()
{

    // Use sizeof() to know size the data types
    printf("The size of int: %d\n", sizeof(int));
    printf("The size of char: %d\n", sizeof(char));
    printf("The size of float: %d\n", sizeof(float));
    printf("The size of double: %d", sizeof(double));

    return 0;
}
```

Pointer Data Types in C

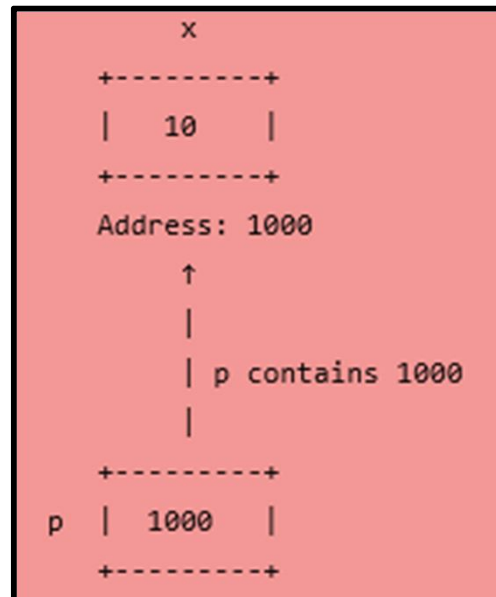
- A **pointer** is a **variable** that stores the **memory address** of **another variable**. General syntax:

data_type *pointer_name;

There are two important operators:

Operator	Meaning	Example
&	Address-of operator	&x gives address of x
*	Dereference operator	*p gives value stored at the address in p

- Suppose: **int x = 10;**
int *p = &x;



Pointer Data Types in C

```
#include <stdio.h>
int main()
{
    int x = 10;
    int *p;
    p = &x;
    printf("Value of x = %d\n", x);
    printf("Address of x = %p\n", &x);
    printf("Value stored in p = %p\n", p);
    printf("Value pointed by p = %d\n", *p);
    return 0;
}
```

```
Value of x = 10
Address of x = 0x7ffef1206384
Value stored in p = 0x7ffef1206384
Value pointed by p = 10
```

Type Conversion in C

- Process of changing **one data type into another**.
- This can happen **automatically** by the **compiler** or **manually**.
- **Only performed** between data types where such a **conversion is possible**.

There are two types of conversion in C:

- ❖ **Implicit Conversion** (**automatically**)
- ❖ **Explicit Conversion** (**manually**)

Escape Sequence in C

- An **escape sequence** is a special combination of characters in C that begins with a **backslash** (****).
- Used to represent **special characters** or perform **special actions** in a **string** / **character constant**.

Escape Sequence	Meaning	Example
\n	New line	<code>printf("Hello\nWorld");</code>
\t	Horizontal tab	<code>printf("Name\tAge");</code>
\'	Single quotation mark	<code>printf("\'");</code>
\"	Double quotation mark	<code>printf("\"Hello\"");</code>
\0	Null character	Used to terminate a string

Type Conversion in C

- *In order to effectively develop C programs, it will be necessary to understand the rules that are used for the **implicit conversion** of floating point and integer values in C.*
 - (a) *An operation between **an integer** and **integer** always yields in an **integer result**.*
 - (b) *An operation between **a real** and **real** always yields a **real result**.*
 - (c) *An operation between **an integer** and **real** always yields a **real result**.*

Type Conversion in C

Assume that **k** is an **integer** variable and **a** is a **real** variable.

Arithmetic Instruction	Result	Arithmetic Instruction	Result
$k = 2 / 9$	0	$a = 2 / 9$	0.0
$k = 2.0 / 9$	0	$a = 2.0 / 9$	0.222222
$k = 2 / 9.0$	0	$a = 2 / 9.0$	0.222222
$k = 2.0 / 9.0$	0	$a = 2.0 / 9.0$	0.222222
$k = 9 / 2$	4	$a = 9 / 2$	4.0
$k = 9.0 / 2$	4	$a = 9.0 / 2$	4.5
$k = 9 / 2.0$	4	$a = 9 / 2.0$	4.5
$k = 9.0 / 2.0$	4	$a = 9.0 / 2.0$	4.5

Type Conversion in C

- **Implicit Conversion:**

*Implicit conversion is done **automatically** by the compiler when you assign **a value of one type to another**.*

```
#include <stdio.h>
int main()
{
    float num1 = 9;
    int num2 = 3.245;
    float sum;
    printf("%f\n", num1);
    printf("%d\n", num2);
    sum = 5 / 2;
    printf("%f", sum);
    return 0;
}
```

```
9.000000
3
2.000000

...Program finished with exit code 0
Press ENTER to exit console.
```

Type Conversion in C

- **Explicit Conversion:**

Explicit conversion is **done manually** by placing the **type in parentheses ()** in front of the value.

```
#include <stdio.h>
int main()
{
    int num1 = 5, num2 = 2;
    float sum;
    sum = num1 / num2;
    printf("%f", sum);
    sum = (float) num1 / num2;
    printf("\n%f", sum);
    return 0;
}
```

```
2.000000
2.500000

...Program finished with exit code 0
Press ENTER to exit console. □
```

Data Types in C

Data Type	Modifiers	Size (bytes)	Range
char	signed	1	-128 to 127
	unsigned (default)	1	0 to 255
short int	signed (default)	2	-32,768 to 32,767
	unsigned	2	0 to 65,535
int	signed (default)	4	-2,147,483,648 to 2,147,483,647
	unsigned	4	0 to 4,294,967,295
long int	signed (default)	8	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
	unsigned	8	0 to 18,446,744,073,709,551,615
double	None	8	~1.7E-308 to 1.7E+308
long double	None	16	Higher precision, range varies depending on implementation



Thank You!

